

Šimon Hykl

Aircraft Design | Thermal Management

Location: Prague, Czech Republic
(committed to relocation)

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Aeronautical Engineer

I am a recent graduate of aeronautical engineering at CTU Prague with a strong foundation in electrical engineering and additional academic experience at Delft University of Technology. Specialized in the design of classic small aircraft and compliance with European airworthiness standards (UL-2, CS-22, CS-23). 6 years of Formula Student experience with motorsport-focused mechanical design, thermal management, and CFD. Used to high-pressure teamwork in a fast-paced environment. Committed to international relocation.

Education

Czech Technical University in Prague, Faculty of Electrical Engineering <i>Bachelor's degree (B.Eng) in electrical engineering</i>	Sep 2019 – Jun 2023
Study program: Open Electronic Systems Thesis: Scattering of an Electromagnetic Wave on a Two-Dimensionally Periodic Structure Grade: B	
Czech Technical University in Prague, Faculty of Mechanical Engineering <i>Master's degree (M.Eng) in aerospace engineering</i>	Sep 2023 – Sep 2026
Study program: Aircraft & Unmanned Systems Thesis: Design of a Cooling System for a Formula Student Traction Accumulator Grade: A	
Delft University of Technology <i>ERASMUS+ (1 semester)</i>	Sep 2025 – Jan 2026
Aircraft avionics and operations, Advanced heat transfer, Non-linear dynamics, Design of Lightweight Structures, Multiphysics simulations	

Experience (Formula Student)

eForce Prague Formula <i>Electrical Engineer, Mechanical Engineer</i>	2019 – 2025
PCB hardware design and testing (initially), Design and integration of composite structures, Mechanical design of high-voltage batteries, Space-constrained mechanical layout, Battery thermal management, Cooling design, CFD, FEA, Data analysis Successful development of Czechia's first self-driving Formula Student car (2022–2023) Driver (2023) Leader of the LV electronics department (2025) Leader of the traction battery subgroup (2025) Representation of the team at EXPO 2025 in Osaka, Japan The team's world ranking: 7 (DV), 12 (EV)	

Technical Skills

Engineering Analysis	: Internal and external CFD, Conjugate heat transfer, Experimental CFD validation, Thermo-electrical multiphysics, Data-driven cooling design, Design of experiments, Cost evaluation, Genetic algorithms, Neural networks
Simulation	: Ansys Fluent, Ansys Mechanical, Ansys SpaceClaim, COMSOL Multiphysics
CAD & Packaging	: Dassault 3DEXperience, CATIA V5, Autodesk Fusion 360, High-Density Mechanical Packaging
Electronics & Hardware	: Altium Designer, KiCAD, High-Voltage Traction Batteries, PCB Design and Testing
Programming & Tools	: MATLAB, C (basics), LaTeX, Git
Languages	: Czech (native), English (fluent – CEFR C2), German (intermediate), Japanese (elementary)